

Chapter 4 — Elements of Data Processing

Practice Test: 50 Multiple Choice Questions (MCQs)

1. The process of producing a set of instructions called a program is known as:
 - A) Processing
 - B) Programming
 - C) Debugging
 - D) Flowcharting
2. A program makes a computer perform a:
 - A) Random activity
 - B) Specific activity
 - C) Physical activity
 - D) Manual activity
3. Computers do not solve problems by themselves; they:
 - A) Make decisions independently
 - B) Carry out instructions
 - C) Create programs automatically
 - D) Change the input
4. Who decides what steps the computer should take?
 - A) Computer
 - B) Programmer
 - C) User only
 - D) Compiler
5. The form of input and output is specified by the:
 - A) Programmer
 - B) Computer
 - C) Printer
 - D) Interpreter
6. A plan for solving a problem is coded in a:
 - A) Machine
 - B) Programming language
 - C) Database
 - D) Flowchart only
7. A program is translated into machine code by a:
 - A) Keyboard
 - B) Translating program
 - C) Monitor
 - D) Printer
8. Which of the following is a translating program?
 - A) Compiler
 - B) Interpreter
 - C) Assembler
 - D) All of these
9. After translation into machine code, the program is ready to be:
 - A) Deleted
 - B) Executed
 - C) Printed

D) Designed

10. During execution, the steps in the program are:

- A) Ignored
- B) Carried out
- C) Deleted
- D) Rewritten

11. The first step in the evaluation from problem to program solution is:

- A) Coding
- B) Defining the problem
- C) Debugging
- D) Final output

12. The second step in the programming process is:

- A) Analysing the problem
- B) Coding
- C) Processing
- D) Testing

13. Which comes after analysing the problem?

- A) Final output
- B) Algorithm
- C) Debugging
- D) Coding

14. Flowcharting comes after:

- A) Algorithm
- B) Final output
- C) Debugging
- D) Processing

15. Coding means:

- A) Testing the computer
- B) Programming the problem
- C) Defining the problem
- D) Producing output

16. Which is the sixth stage of the programming process?

- A) Coding
- B) Processing the problem
- C) Algorithm
- D) Debugging

17. Debugging and testing the program is the:

- A) 3rd stage
- B) 5th stage
- C) 7th stage
- D) 8th stage

18. The final stage of the programming process is:

- A) Final output
- B) Coding
- C) Algorithm
- D) Flowcharting

19. Problem definition and program design generally receive what percentage of development time?

- A) 20%
- B) 30%
- C) 40%
- D) 50%

20. Program coding generally receives:

- A) 10%
- B) 20%
- C) 40%
- D) 60%

21. Program debugging, testing and documentation receive:

- A) 20%
- B) 30%
- C) 40%
- D) 50%

22. Total development time allocated in the given distribution is:

- A) 80%
- B) 90%
- C) 100%
- D) 120%

23. Before any other steps can be taken, one must:

- A) Code the program
- B) Define the problem
- C) Debug the program
- D) Execute the program

24. Defining the problem means studying it to:

- A) Change it
- B) Understand it completely
- C) Delete it
- D) Complicate it

25. Which question helps in defining a problem?

- A) Does problem solving require a computer?
- B) What is the computer's color?
- C) Who owns the computer?
- D) What is the computer's price?

26. While defining a problem, we determine whether the available computer is:

- A) Expensive
- B) Capable of solving the problem
- C) New
- D) Portable

27. Which of the following must be decided while defining the problem?

- A) Programming language
- B) Computer color
- C) Keyboard size
- D) Monitor brand

28. The inputs and expected outputs should be identified during:

- A) Defining the problem
- B) Debugging
- C) Final output

D) Execution

29. "How must data be manipulated to produce the desired output?" is a question related to:

- A) Defining the problem
- B) Printing
- C) Hardware installation
- D) Debugging only

30. The process of understanding the problem is called:

- A) Programming
- B) Problem analysis
- C) Coding
- D) Execution

31. The goal of problem analysis is the formal and logical presentation of the problem for:

- A) Manual solution
- B) Computer solution
- C) Printing
- D) Drawing

32. Analysis of a problem requires the programmer to recognize the:

- A) Problem
- B) Printer
- C) Monitor
- D) Keyboard

33. During problem analysis, the programmer identifies inputs and:

- A) Printers
- B) Variables
- C) Monitors
- D) Files only

34. The programmer should list the desired:

- A) Inputs only
- B) Outputs
- C) Hardware
- D) Errors

35. A precise problem statement is formulated during:

- A) Problem analysis
- B) Final output
- C) Coding only
- D) Execution

36. During problem analysis, it is determined whether a _____ is needed for the solution.

- A) Printer
- B) Computer
- C) Keyboard
- D) Scanner

37. During problem analysis, all elements in the problem are determined:

- A) Randomly
- B) Precisely
- C) Approximately
- D) Secretly

38. Input and output conditions should be:

- A) Ignored
- B) Specified
- C) Changed randomly
- D) Deleted

39. Elements to be processed are converted into:

- A) Qualitative terms
- B) Quantitative terms
- C) Graphic terms
- D) Verbal terms

40. The operations and manipulations required to produce desired results must be determined as:

- A) Pictures
- B) Steps
- C) Errors
- D) Outputs

41. In problem analysis, the _____ of steps is ascertained.

- A) Cost
- B) Sequence
- C) Color
- D) Size

42. Relationships among all elements entering into the problem's solution are:

- A) Ignored
- B) Established
- C) Deleted
- D) Printed

43. The word algorithm originates from the word:

- A) Algebra
- B) Algorism
- C) Algorithmic
- D) Arithmetic

44. Algorism means:

- A) Art of programming
- B) Art of computing with Arabic numerals
- C) Art of drawing
- D) Art of coding

45. The word algorithm is in honour of:

- A) Isaac Newton
- B) Abu Jafar Muhammad Ibn Musa Al-Khwarizmi
- C) Albert Einstein
- D) Aristotle

46. Al-Khwarizmi was a famous:

- A) Computer scientist
- B) Arab mathematician
- C) Physicist
- D) Programmer

47. An algorithm for solving a problem is a procedure that consists of:

- A) Infinite steps
- B) Finite number of discrete steps
- C) Random steps

D) No steps

48. An algorithm's steps must be performed in a:

- A) Specified order
- B) Random order
- C) Changing order
- D) Reverse order only

49. An algorithm should eventually reach a solution or report that:

- A) The computer is off
- B) There is no solution
- C) The program is expensive
- D) Input is large

50. Which of the following is NOT a property of an algorithm?

- A) Finiteness
- B) Definiteness
- C) Completeness
- D) Entertainment

Answer Key

1. B	2. B	3. B	4. B	5. A
6. B	7. B	8. D	9. B	10. B
11. B	12. A	13. B	14. A	15. B
16. B	17. C	18. A	19. C	20. B
21. C	22. C	23. B	24. B	25. A
26. B	27. A	28. A	29. A	30. B
31. B	32. A	33. B	34. B	35. A
36. B	37. B	38. B	39. B	40. B
41. B	42. B	43. B	44. B	45. B
46. B	47. B	48. A	49. B	50. D

Important Algorithm Properties:

- **Finiteness:** An algorithm must terminate after a finite number of steps.
- **Definiteness:** All steps must be precisely defined and unambiguous.
- **Completeness:** The rules must be complete so the algorithm can solve the problems of its intended class.
- **Efficiency:** An algorithm can be efficient or inefficient.